

VSLA Empirical Replication Report

Impact of Village Savings and Loan Associations: Evidence from a Cluster Randomized Trial

Empirical Replication Project

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GitHub Repository: <https://github.com/desiayuni/fiscal-decentralization-poverty-indonesia>

1. Research Question

The impact of Village Savings and Loan Associations (VSLAs), cluster randomized trial (RCT); replication of treatment effects using STATA; robustness with additional household controls.

2. Project Objectives

- Replicate core results in "Impact of Village Savings and Loan Associations: Evidence from a Cluster Randomized Trial" (J Dev Econ, 2016)
- Illustrate basic applied micro econometric methods and evaluate the efficacy of VSLA participation in increasing household loan access
- Generate baseline and robustness results including demographic household controls

3. Data

- Household-level cluster RCT data with treatment / control variables, village clustering, post-treatment outcomes
- Dependent variable: anyloan (household loan access)
- Key independent variable: treat (village VSLA participation)
- Key robustness variables: fhead (female-headed household), headage, headedu (education level)

4. Methodology

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- OLS regression with clustered standard errors; block fixed effects
 - Baseline: impact of VSLAs on loan access
 - Robustness test: inclusion of demographic controls; robustness check
 - Methods employed: OLS, clustered SE, fixed effects, robustness checks, comparative specifications, coefficient plotting

5. Baseline results

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- Effect size ≈ 0.104 ; statistically significant at 1%
 - Interpretation: VSLA participation positively impacts loan access; consistent with concept of financial inclusion through community-based finance

Table 1. Baseline and Robustness Estimates

	Baseline Anyloan	With Controls Anyloan
Treat	0.104*** (0.024)	0.0973*** (0.022)
Large fishing	0 (.)	0 (.)
Other fishing	-0.028 (0.046)	-0.016 (0.044)
Large non-fishing	-0.087** (0.043)	-0.092** (0.042)
Eager	0.008 (0.047)	0.010 (0.047)
With rice irrigation scheme	0.029 (0.049)	0.016 (0.055)
With World Vision Project	0.043 (0.054)	0.062 (0.052)
Remaining	-0.008 (0.046)	-0.003 (0.044)
Fhead		0.058** (0.029)

Headage		0.000772 (0.000807)
Headed		0.015*** (0.004)
_cons	0.136*** (0.041)	-0.008 (0.061)
<i>N</i>	1636	1614

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

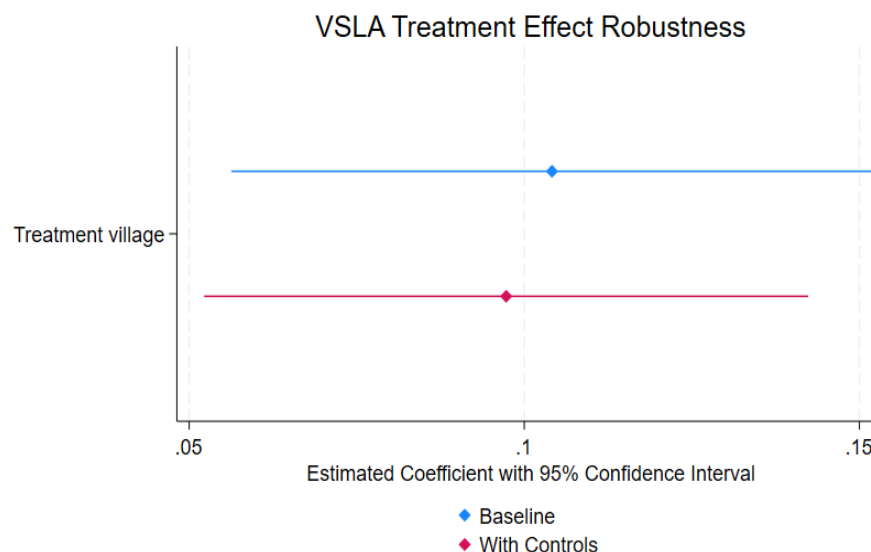
Notes: Standard errors clustered at the village level. Block fixed effects included in all specifications.

The treatment coefficient remains positive and statistically significant across specifications. After including demographic household controls, the estimated treatment effect decreases slightly from approximately 0.104 to 0.097, suggesting that the main findings are relatively robust to observable household characteristics.

6. Robustness results

- Including household characteristics: effect size ≈ 0.097 ; still statistically significant at 1%
- Interpretation: Effect robust to observable household demographics; standard applied micro econometrics method

7. Empirical Visualization



- Publication-quality coefficient plots illustrating positive, statistically significant effect across specifications

Figure 1 illustrates that the estimated treatment effect remains positive and statistically significant across both the baseline and controlled specifications. Confidence intervals remain above zero, supporting the robustness of the estimated treatment effect.

8. Workflow and software used

- Preparation, processing, generating new variables; estimating treatment effect; robustness testing; creating tables, figures; documenting methods and results
- Software used: STATA, GitHub, Markdown

9. Skills illustrated

- Applied econometric techniques, STATA workflow, clustered standard errors, comparing specifications, robustness testing, replication, visualizing results, reproducibility

10. Conclusions

- Replicates core findings of an RCT; VSLAs positively and significantly improve loan access; robust to demographic household controls
- Illustrates basic applied reproducibility practices prevalent in development economics literature

11. GitHub repository

- All replication files (STATA scripts, outputs, robustness analysis, visualizations) uploaded to GitHub